

Documentation chemical data input interfaz_remix

This document contains the documentation for the chemical data input in the code **interfaz_remix**. There are maximum 6 files to be read. The minimum of input files is 4 (the ones that end in “_quim_loc.dat”, “_sist_quim.dat”, “_tar_wat.dat” and “CV_params.dat”). The other two (ending in “_tar_sol.dat” and “_tar_gas.dat”) are optional, depending if the chemical system contains solids and/or gases. All of the files must contain the **same root** in their names, and they must live in the **same directory**.

We describe the input files with an example of two waters that mix: an inflow water in equilibrium with atmospheric O_2 and CO_2 and a resident water in equilibrium with calcite. This example can be found in the directory “..\examples\calcite_eq”.

Chemical system file: “_sist_quim.dat”

This file contains the information related to the chemical system (species, reactions, phases). It is read by the **Chemical System** class.

- **Line 1:** Title of the problem
- **Line 2:** Section separation
- **Line 3:** Section name
 - If Section name = ‘PRIMARY AQUEOUS SPECIES’ or ‘AQUEOUS COMPLEXES’:
 - i. **Line 3a:** *sp_name* (string)
 - i. Name of species in database “*master25_modif.dat*”
 - ii. **Line 3b:** end of section
 - If Section name = ‘MINERALS’ or ‘GASES’:
 - i. **Line 3a:** *sp_name* (string), *flag_eq* (logical), *flag_cst_act* (logical)
 - i. *sp_name*: Name of species in database “*master25_modif.dat*”
 - ii. *flag_eq*: TRUE if equilibrium reaction, FALSE otherwise
 - iii. *flag_cst_act*: TRUE if species has constant activity, FALSE otherwise
 - ii. **Line 3b:** end of section
 - If Section name = ‘REDOX REACTIONS’:
 - i. **Line 3a:** *react_name* (string), *flag_eq* (logical)
 - i. *react_name*: Name of redox reaction in database “*Monod_DB.dat*”
 - ii. *flag_eq*: TRUE if equilibrium reaction, FALSE otherwise
 - ii. **Line 3b:** end of section
 - If Section name = ‘SURFACE COMPLEXES’:
 - i. **Line 3a:** *sp_name* (string)
 - i. *sp_name*: Name of species in database “*master25_modif.dat*”
 - ii. **Line 3b:** end of section
- **Line 4:** Section separation
- **Line 5:** end of file

'TITLE OF THE PROBLEM: Calcite in equilibrium'

'-----'

'PRIMARY AQUEOUS SPECIES'

'h2o(p)' ! species name

'h+'

'ca+2'

'o2(aq)'

'hco3-'

'*'

'MINERALS'

'calcite' .true. .true. ! name of mineral, flag equilibrium, flag constant activity

'*'.true. .true.

'GASES'

'o2(g)' .true. .true. ! name of gas, flag equilibrium, flag constant activity

'co2(g)' .true. .true.

'*'.true. .true.

'-----'

'end'

Local chemistry file: “_quim_loc.dat”

This file contains the **local** chemical information (water types, mineral zones, gas zones, surfaces). It is read by the **Chemistry** class.

- **Line 1:** Name of the problem
- **Line 2:** Section separation
- **Section name:** ‘INITIAL AND BOUNDARY WATER TYPES’
 - **Line 1:** *act_coeffs_model* (integer)
 - 0. Ideal
 - 1. Debye-Hückel
 - 2. Debye-Hückel extended
 - 3. Davies
 - 4. Truesdell-Jones
 - **Line 2:** *num_wat_types* (integer)
 - i. Number of water types (includes initial, recharge, boundary, etc)
 - **Line 3:** *ind_wat_type* (integer), *temp* (real)
 - i. *ind_wat_type*: index of water type
 - ii. *temp*: temperature of solution (in Celsius)
 - **Line 3a:** *name* (string)
 - i. Name of water type
 - **Line 3b:** water type data begins
 - i. *icon*: initial condition type
 - 1. prescribed primary concentration
 - 2. prescribed total concentration
 - 3. prescribed activity
 - 4. primary species is in equilibrium with a constrain (mineral, gas or aqueous complex)
 - ii. *guess*: Initial guess for primary species concentration
 - iii. *ctot*: (all concentrations are expressed in molalities)
 - i. *icon=1*: primary species concentration
 - ii. *icon=2*: total concentration of primary species
 - iii. *icon=3*: activity of primary species
 - iv. *icon=4*: activity of constrain that is in equilibrium with this primary species
 - iv. *constrain*: if *icon=4*, it is the species in equilibrium with the primary species
 - **Line 3c:** *prim_sp_name* (string), *icon* (integer), *guess* (real), *ctot* (real), *constrain* (string)
 - **Line 3d:** end of water type data
 - **Line 4:** end of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'

'-----'

'INITIAL AND BOUNDARY WATER TYPES'

0                                ! activity coefficients model

2                                ! number of water types

1  25.0                          ! index water type, temperature (C)

'boundary'                      ! name of water type

'   icon    guess    ctot    constrain'

'h2o(p)'  3      1d0      1d0          ""

'ca+2'    1      1d-9      1d-9          ""

'o2(aq)'  4      1d-9      2d-1      'o2(g)'

'hco3-'   4      1d-9      3.5d-4      'co2(g)'

'h+'      1      1d-7      1d-7          ""

'*'       0      0.0      0.0          ""

2  25.0                          ! index water type, temperature (C)

'resident'                      ! name of water type

'   icon    guess    ctot    constrain'

'h2o(p)'  3      1d0      1d0          ""

'ca+2'    1      1d-3      1d-3          ""

'o2(aq)'  1      1d-6      1d-6          ""

'hco3-'   4      1d-9      1d0          'calcite'

'h+'      1      1d-5      1d-5          ""

'*'       0      0.0      0.0          ""

'-----'
```

- Section name: 'INITIAL MINERAL ZONES'
 - **Line 1:** *num_min_zones* (integer)
 - i. Number of mineral zones
 - **Line 2a:** *ind_min_zone* (integer), *temp* (real)
 - i. *ind_min_zone*: index of mineral zone
 - ii. *temp*: temperature of mineral zone (in Celsius)
 - **Line 2b:** *name* (string)
 - i. Name of mineral zone
 - **Line 2c:** mineral zone data begins
 - **Line 2d:** *min_name* (string), *vol_frac* (real), *react_surf* (real)
 - i. *min_name*: name of mineral in mineral zone (must be in the chemical system)
 - ii. *vol_frac*: initial volumetric fraction of mineral
 - iii. *react_surf*: specific surface of mineral (surface of mineral per unit volume of rock)
 - **Line 2e:** end of mineral zone data
 - **Line 3:** end of section

'INITIAL MINERAL ZONES'

1 ! number of mineral zones

1 25.0 ! index of mineral zone, temperature (C)

'calcite' ! name of mineral zone

'mineral vol.frac. area(m2/m3rock)'

'calcite' 0.50 1d2

'*' 0.00 0.00

'-----'

- Section name: 'INITIAL AND BOUNDARY GAS ZONES'
 - **Line 1:** *num_gas_zones* (integer)
 - i. Number of gas zones
 - **Line 2a:** *ind_gas_zone* (integer), *temp* (real), *vol* (real)
 - i. *ind_gas_zone*: index of gas zone
 - ii. *temp*: temperature of gas zone (in Celsius)
 - iii. *vol*: total volume of gas
 - **Line 2b:** gas zone data begins
 - **Line 2c:** *name* (string)
 - i. Name of gas zone
 - **Line 2d:** *gas_name* (string), *part_press* (real)
 - i. *gas_name*: name of gas in gas zone (must be in the chemical system)
 - ii. *part_press*: initial partial pressure (equivalent to activity) of gas (in atm)
 - **Line 2e:** end of gas zone data
 - **Line 3:** end of section
 - **Line 4:** end of file

'INITIAL AND BOUNDARY GAS ZONES'

1 ! number of gas zones

1 25 1d9 ! index gas zone, temperature (C), volume

'atmosphere' ! name of gas zone

'gas partial pressure'

'o2(g)' 2d-1

'co2(g)' 3.5d-4

'*' 0.00

'-----'

'end'

Waters file: “_tar_wat.dat”

This file contains the number of waters (target, boundary and recharge) and the **association** between waters, target solids and/or target gases. This file is read by the **Chemistry** class.

- **Line 1:** Name of the problem
- **Line 2:** Section separation
- **Section name:** 'WATERS'
 - **Line 1:** *num_wat* (integer)
 - Number of waters
 - **Line 2:** *num_bd_wat* (integer)
 - Number of boundary waters
 - **Line 3:** *num_rech_wat* (integer)
 - Number of recharge waters
 - **Lines 4 - 4+n° intervals:** *int_wat* (string), *iwtype* (integer), *int_sol* (integer), *int_gas*(integer)
 - *int_wat*: Interval of waters
 - *iwtype*: Index water type in section 'INITIAL AND BOUNDARY WATER TYPES'
 - *int_sol*: Interval of target solids associated with *int_wat*
 - *int_gas*: Interval of target gases associated with *int_wat*
 - **The waters must be written in the following order: *target waters*, *boundary waters* and *recharge waters*.**
 - End of section

'TITLE OF THE PROBLEM: Calcite in equilibrium'

'WATERS'

12 ! number of waters

2 ! number of boundary waters

0 ! number of recharge waters

1-10 2 1-10 0 ! waters interval, water type index, target solids interval, target gases interval

11 1 0 0

12 2 0 0

'end'

Target solids file: “_tar_sol.dat”

This file contains the **association** between target solids and solid zones. It is read by the **Chemistry** class. The only solid zones implemented so far are the **mineral** and **exchange** zones. **IMPORTANT**: this file will not be read by the code if there are no solid zones.

- **Line 1**: Name of the problem
- **Line 2**: Section separation
- **Section name**: ‘TARGET SOLIDS’
 - **Line 1**: *num_tar_sol* (integer)
 - Number of target solids
 - **Line 2**: *interval* (string), *iszone* (integer)
 - *interval*: Target solids interval
 - *iszone*: Index of mineral zone in section ‘INITIAL MINERAL ZONES’
 - end of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'

'-----'

'TARGET SOLIDS'

10                                ! number of target solids

1-10    1                        ! target solids interval, solid zone index

'-----'

'end'
```

Target gases file: “_tar_gas.dat”

This file contains the **association** between target gases and gas zones. It is read by the **Chemistry** class. **IMPORTANT**: this file will not be read by the code if there are no gas zones. Note that the "calcite_eq" example does not include this file, since its waters have no associated target gases (the example below is only illustrative).

- **Line 1**: Name of the problem
- **Line 2**: Section separation
- **Section name**: ‘TARGET GASES’
 - **Line 1**: *num_tar_gas* (integer)
 - Number of target gases
 - **Line 2**: *interval* (string), *igzone* (integer)
 - *interval*: Target gases interval
 - *igzone*: Index of gas zone in section ‘INITIAL AND BOUNDARY GAS ZONES’
 - end of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'

'-----'

'TARGET GASES'

1                                ! number of target gases

1      1                        ! target gases interval, gas zone index

'-----'

'end'
```

Convergence parameters file: “_CV_params.dat”

This file contains some **convergence parameters** for solving iterative methods, such as Newton-Raphson for speciation. These parameters are the absolute and relative tolerances, and the damping parameter for estimating reaction rates in downstream waters. The remaining parameters are set by default in the code, such as the maximum number of iterations allowed. This file is read by the **Convergence Parameters** structure.

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'
```

```
'.....'
```

```
'CONVERGENCE PARAMETERS'
```

```
1d-14          ! absolute tolerance
```

```
-14            ! log absolute tolerance
```

```
1d-14          ! relative tolerance
```

```
-14            ! log relative tolerance
```

```
0d0            ! Parameter for estimation of reaction rates
```

```
'.....'
```